PersonaScienceBench: Evaluating Scientific Agents with Diverse Simulated Researchers

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Abstract

TODO: Draft the abstract.

1 Introduction

Scientific work is collaborative. Language-model agents now perform substantial parts of the scientific workflow. They search literature, propose hypotheses, write and execute analysis code, and assemble results into reports (Lu et al., 2024; Yamada et al., 2025; Gottweis et al., 2025; Mitchener et al., 2025b). The benchmarks used to measure this progress have followed the same trajectory, moving from recall and reasoning questions toward executable analysis tasks drawn from real studies (Majumder et al., 2024; Mitchener et al., 2025a; Kon et al., 2025; Ye et al., 2025). Yet in practice these systems function as co-scientists rather than autonomous investigators, and the case has been made that they are not architecturally built for unsupervised discovery (Bisht et al., 2026). A deployed scientific agent sits inside a collaboration: someone designed the experiment, ran the instrument, and holds the context that determines whether an analysis is valid. That context is frequently undocumented. Which plates were processed on a different day, why three samples were dropped, whether a covariate is confounded with batch, which normalization the lab actually applies—these facts live with the researcher, not in the data directory. The agent's scientific output therefore depends not only on its own reasoning but on an exchange with a particular human who may explain readily, answer only what was literally asked, or misremember a detail.

The evaluation gap. Most scientific agent benchmarks specify each task completely and uniformly, which measures capability under ideal information but says nothing about capability under realistic information transfer. A recent line of interactive benchmarks closes part of this gap by requiring agents to recognize underspecification and ask: HiL-Bench measures whether coding agents escalate at the right moments (Trinh et al., 2026), SciConvBench evaluates multi-turn clarification for ill-posed computational science requests (Somasekharan et al., 2026), and work in tool-agent settings evaluates interaction with a simulated user under domain rules (Yao et al., 2024; Chen et al., 2026; Ta et al., 2026). These benchmarks establish that asking is hard. They do not, however, vary *whom* the agent is asking. The simulated collaborator is typically a single cooperative interlocutor with one disclosure policy, so the collaborator is a fixed property of the environment rather than an experimental condition. The closest departures from this pattern vary designed behavioral axes—non-collaborative user simulators degrade tool agents through impatience, digression, and incomplete utterances (Shim et al., 2025), and evolved persona policies induce realistic behavioral variation in otherwise cooperative simulators (Chopra et al., 2026)—but do so in retail, airline, and customer service domains where there is no scientific ground truth to get wrong. The result is that three distinct sources of difficulty remain entangled in current evaluation: the intrinsic difficulty of the scientific task, the capability of the agent, and the character of the collaborator the agent happens to be working with. Reporting a single score against a single default collaborator silently averages over the third.

Our approach. This paper introduces PersonaScienceBench, a benchmark that holds the scientific task fixed and varies the simulated researcher. We convert 200 tasks from five scientific agent benchmarks—DiscoveryBench, BixBench, EXP-Bench, SciConvBench, and ASI-Bench—into collaboration tasks, preserving each source's native deliverable and grading rules rather than reducing them to a common format. An actor agent must produce that native deliverable while interacting with a simulated researcher through a shared review protocol. The researcher is conditioned on a complete

rendered profile drawn from MatrAIx, a population-scale simulated-user infrastructure whose persona records span 1,290 categorical dimensions covering background, psychology, capability, and behavior (Li et al., 2026). Our renderer emits the populated attributes as an explicit attribute–value listing grouped by category, so a condition is a fully auditable set of stated attributes rather than authored prose; a single profile typically contributes on the order of two hundred such attributes. The profile is injected verbatim into the researcher simulator's system prompt only, without truncation or projection onto the task domain, so the actor observes personality solely through conversation. Every task is also run with the identical protocol and the identity block removed, giving a paired persona-off control that separates collaborator variation from task difficulty. Evaluator references stay private throughout, and each task–condition cell is repeated across independent episodes so that persona effects can be distinguished from ordinary run-to-run variation. Because the same task is scored under multiple collaborators with a shared baseline, differences are attributable to the collaborator rather than to task selection.

Upstream validation of this persona corpus establishes that conditioning reliably induces declared behavior—91.5% of trials in a controlled adherence study—but does so in Survey, Chatbot, Web, and App environments over a task library concentrated in commerce, software, finance, and healthcare (Li et al., 2026). Scientific research collaboration is not among those environments, and product feedback rather than scientific correctness is the measured outcome. PersonaScienceBench supplies that missing environment and asks the question it enables: whether persona-induced behavioral variation propagates into graded scientific error.

What we measure. Evaluation is deliberately two-sided. Outcome metrics use each source benchmark's native grader, so a persona effect registers as a change in scientific correctness and not merely as a change in conversational tone. Interaction metrics—review rounds, revisions, questions asked, clarification coverage, message length, and language adaptation—are recorded separately, because a collaborator can change an agent's behavior substantially without changing its score, and can change its score without changing observable behavior. Keeping these separate is what allows a failure under a difficult collaborator to be diagnosed as a failure to obtain available information rather than assumed to be one.

Contributions. **TODO:** Draft the contributions.

Section 2 positions this work against scientific agent benchmarks, interactive and simulated-user evaluation, and persona simulation. Section 3 defines the benchmark, Section 4 the experimental protocol, Section 5 the results, and Section 6 mechanism evidence from conversation traces. Section 7 states what simulated collaborators and an outcome-selected panel cannot establish.

2 Related Work

Language Models with Personas. Conditioning language models on persona descriptions can produce varied responses and simulated user behavior (Argyle et al., 2022; Ge et al., 2024). Recent work develops persona generators for user coverage and applies persona-based simulation to interactive-system evaluation (Paglieri et al., 2026; Liu et al., 2026). Non-collaborative user simulators and Persona Policies further show that varying user behavior can change tool-agent performance in service domains (Shim et al., 2025; Chopra et al., 2026). MatrAIx provides structured profiles spanning background, capability, and behavior (Li et al., 2026). We use these profiles to study simulated scientific collaborators, comparing persona-conditioned interactions with independent persona-off repeats to distinguish profile-induced diversity from ordinary run variation.

Agents for Scientific Discovery. Language-model agents increasingly support hypothesis generation, experimental implementation, and data analysis, as illustrated by the AI Scientist, AI co-scientist, and Kosmos (Lu et al., 2024; Yamada et al., 2025; Gottweis et al., 2025; Mitchener et al., 2025b). Benchmarks evaluate these capabilities through data-driven discovery and computational biology (Majumder et al., 2024; Mitchener et al., 2025a), experimental design and execution (Kon et al., 2025), and scientific implementation under varying methodological guidance (Zhou et al., 2026). SciConvBench extends evaluation to multi-turn clarification of underspecified computational-science requests (Somasekharan et al., 2026). These benchmarks assess scientific task performance and information recovery, but generally leave collaborator behavior fixed. PersonaScienceBench builds on their task settings by varying the simulated researcher on matched scientific tasks and measuring interaction patterns alongside source-specific outcomes. This design tests how collaborator variation affects both the scientific process and its results.

3 PersonaScienceBench

3.1 Persona Construction

TODO: Divid: Introduce persona construction and the relevant background from MatrAIx (Li et al., 2026), then describe how its profiles are used to construct the simulated researchers in this benchmark.

3.2 Dataset Construction for Interactive Scientific Discovery

We construct a dataset of 200 scientific tasks, with 40 tasks from each of five benchmarks: DiscoveryBench (Majumder et al., 2024), BixBench (Mitchener et al., 2025a), EXP-Bench (Kon et al., 2025), SciConvBench (Somasekharan et al., 2026), and ASI-Bench (Zhou et al., 2026). Together, they cover data analysis, experimental design, scientific implementation, and specification clarification across four scientific domains. Table 1 shows the distribution by benchmark and domain, alongside the generic episode in Algorithm 1. Equal task counts balance the source benchmarks, while domain coverage reflects the subject matter available in each source.

Table 1: Task counts by benchmark and domain.

Benchmark	Life	Phys./ Eng.	Math./ CS	Soc./ Hum.	Total
DiscoveryBench	5	0	3	32	40
BixBench	40	0	0	0	40
EXP-Bench	0	6	34	0	40
SciConvBench	0	34	6	0	40
ASI-Bench	5	23	12	0	40
Total	**50**	**63**	**55**	**32**	**200**

Domains: Life, life sciences; Phys./Eng., physical sciences and engineering; Math./CS, mathematics and computer science; Soc./Hum., social sciences and humanities.

Each task is converted into an interactive episode with separate information for the actor, researcher, and evaluator.

Algorithm 1: Generic benchmark episode.

Require: Source task t; persona p (or persona off)
$(b, k, e, \pi) \leftarrow \text{Convert}(t)$
Initialize actor A with public inputs b.
Initialize simulator S with context k and p.
$h \leftarrow []$ {shared conversation}
while protocol π permits another turn **do**
  if π assigns the next turn to A **then**
    $m \leftarrow A.\text{ActAndReport}(h)$
  else
    $m \leftarrow S.\text{Respond}(h)$
  end if
  Append m to h.
end while
$y \leftarrow \text{SealSubmission}(A, h)$
return $e.\text{Evaluate}(y, h)$

Conversion separates public inputs b, researcher context k, and private evaluator e, and selects protocol π. Actor actions use only the tools permitted by the task; evaluator references stay private.

The actor receives the scientific request, permitted input files, and output requirements. The researcher receives source-attributed context, fixed requirements, or optional methodological advice, together with the shared conversation. Reference answers, solution artifacts, and hidden grading information remain reserved for the evaluator. Researcher context is drawn from permitted task material rather than generated from reference answers. Their role differs across sources: BixBench schema information is also discoverable from public files, whereas SciConvBench's intended specification is held by the researcher. Interaction therefore provides both consultation and access to withheld context, depending on the task. The scientific requirements and available task context remain fixed across persona conditions.

Conversion retains source-specific deliverables and scoring criteria, with explicit adaptations to support interaction. EXP-Bench is converted to a design-only track, so its scores concern proposed experiments rather than executed research. For ASI-Bench, the actor receives the objective-only prompt and the researcher receives eligible methodological guidance from the next-richer prompt level. This creates an execution task with consultation available; it is not the source's autonomous objective-only condition. These adaptations support comparisons among collaborators within our benchmark rather than direct comparisons with the original leaderboard scores.

DiscoveryBench, BixBench, EXP-Bench, and ASI-Bench share an interaction protocol with approach and result review. The actor inspects the inputs and proposes an approach; the researcher approves it or requests clarification. The actor then performs the analysis or prepares the design and submits a result for review. The researcher can request a revision or conclude the episode, after which the controller seals the deliverable for separate evaluation. The researcher sees actor-shared evidence rather than private tool logs. SciConvBench retains its native clarification dialogue: the actor asks questions to recover or correct the specification, without executing a scientific solver. Persona profiles condition

only the researcher; the actor encounters their effects through the conversation.

Task selection admits numerical scientific workflows and diagnostic plots while excluding image-centric objectives such as microscopy interpretation, computer vision, and visual generation. The same tasks and permitted inputs are used across persona conditions, allowing collaborator effects to be compared on a common scientific task set.

4 Experimental Setup

TODO: Describe the experimental setup.

5 Results

TODO: Add results and figures.

6 Discussion and Conversation Case Studies

TODO: Draft the discussion and conversation case studies.

7 Limitations and Responsible Use

TODO: Document limitations and responsible use.

8 Conclusion

TODO: Draft the conclusion.

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A Supplementary Material

TODO: Add supplementary material.